The above figure shows the pdfs of two continuous random variables whose curves are bell-shaped with same mean but different variances.

Example 11.16

Suppose that $f(x)$ given below represents a probability mass function,

x	1	2	3	4	5	6
$f(x)$	c^2	$2c^2$	$3c^2$	$4c^2$	c	$2c$

Find (i) the value of c (ii) Mean and variance.

Solution

(i) Since $f(x)$ is a probability mass function, $f(x) \geq 0$ for all x, and $\sum_x f(x) = 1$.

Thus, $\sum_x f(x) = 1$

$$c^2 + 2c^2 + 3c^2 + 4c^2 + c + 2c = 1$$

$$c = \frac{1}{5} \text{ or } -\frac{1}{2}.$$

Since $f(x) \geq 0$ for all x, the possible value of c is $\frac{1}{5}$.

Hence, the probability mass function is

x	1	2	3	4	5	6
$f(x)$	$\frac{1}{25}$	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{4}{25}$	$\frac{1}{5}$	$\frac{2}{5}$

(ii) To find mean and variance, let us use the following table

x	$f(x)$	$x\,f(x)$	$x^2 f(x)$
1	$\frac{1}{25}$	$\frac{1}{25}$	$\frac{1}{25}$
2	$\frac{2}{25}$	$\frac{4}{25}$	$\frac{8}{25}$
3	$\frac{3}{25}$	$\frac{9}{25}$	$\frac{27}{25}$
4	$\frac{4}{25}$	$\frac{16}{25}$	$\frac{64}{25}$
5	$\frac{1}{5}$	$\frac{5}{5}$	$\frac{25}{5}$
6	$\frac{2}{5}$	$\frac{12}{5}$	$\frac{72}{5}$
	$\sum f(x) = 1$	$\sum x\,f(x) = \frac{115}{25}$	$\sum x^2 f(x) = \frac{585}{25}$

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